

SparkSNMP

SparkSNMP is an Ignition device driver that polls network gear, UPSes, and other SNMP-capable devices as OPC UA device connections. It supports SNMP v1, v2c, and v3, and needs no MIB compilation step to browse or address a device.

Version 0.1.2 Ignition 8.1.19+ and 8.3 Runs on: any gateway

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1. Overview

SparkSNMP is a standalone device driver. It does not require any other SparkBridge module; it appears in Ignition as an OPC UA device type and is licensed separately, sold and kept per gateway.

2. What You Need

- Ignition 8.1.19 or newer, or the 8.3 line. One bundle per line.
- An SNMP agent reachable over UDP, default agent port 161.
- For SNMPv3: USM auth MD5/SHA1/SHA224/SHA256/SHA384/SHA512, and privacy DES/AES128/AES192/AES256, on the read path.

3. Install

Download the signed release build and install it through Config > Modules > Install or Upgrade a Module, the same as any SparkBridge module; no build step is required.

4. License

Licensed per gateway, bought once and kept. An unlicensed gateway runs identically to a licensed one; drop a license key at `<Ignition data dir>/sparkbridge-license.key`.

From release 2.7.0, every SparkBridge module runs on Ignition's standard two-hour module trial (the gateway's Reset Trial button restarts the clock). A verified key overrides the trial entirely and the trial clock is then ignored. Drop the key at `<data>/sparkbridge-license.key`; each module re-checks the file every 30 seconds, so installing, replacing, or removing a key takes effect with no restart.

5. Configure

Config > OPC UA > Device Connections > Create new Device > the SNMP device type. Select the SNMP version (V1/V2C/V3), set Community (V1/V2C) or the USM user/auth/privacy settings (V3), and the agent host and UDP port.

Setting	Meaning, default
Community	Read community string for V1/V2C, for example <code>public</code> . A separate read-write community enables SET.
Allow SNMP SET Writes	Must be ticked, plus a write-capable community, before OPC UA writes reach the agent.

Addressing (OPC item paths):

Form	Meaning
<code>1.3.6.1.2.1.1.1.0</code>	Raw instance OID.
<code>sysDescr.0 / sysDescr</code>	MIB-2 scalar by name, .0 implied.
<code>ifDescr[1]</code>	Table column plus row index.
<code>Interfaces/ifTable/1/ifDescr</code>	Browse-tree path.

Append `|string`, `|hex`, `|bytes`, or `|seconds` to force presentation, for example `ifPhysAddress[1]|hex`.

Custom vendor MIBs upload under Config > SparkSNMP > Custom MIBs, or via `POST /data/com.greenguardusa.sparksnmp/mibs?name=<module>` with the MIB text as the body.

6. Operate

Day to day, SparkSNMP polls its configured OIDs and tables like any OPC UA device. Watch `GET status.json` under the device's route group for lane health, and use the OPC UA browser to confirm MIB-2 folders (`sysDescr`, `ifTable`, `ipAddrTable`) and any uploaded vendor subtrees are populated.

7. Verify

- Check `GET status.json` under the device's route group; a healthy lane reports `RUNNING`.
- Browse the device in an OPC UA client and confirm MIB-2 folders (`sysDescr`, `ifTable`, `ipAddrTable`) and any uploaded vendor subtrees appear.

8. Troubleshoot

Symptom	Cause / fix
Writes never reach the agent	Allow SNMP SET Writes must be ticked and the configured community must be write-capable.
Custom MIB names do not resolve	Confirm the MIB was uploaded under Config > SparkSNMP > Custom MIBs or via the <code>/mibs</code> endpoint before addressing it by name.

9. Reference

Module id	<code>com.greenguardusa.sparksnmp</code>
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Default agent port UDP 161

Custom MIB upload endpoint POST /data/com.greenguardusa.sparksnmp/mibs?name=<module>

License file <Ignition data dir>/sparkbridge-license.key

10. Support

Docs page section: </sparkbridge/docs#sparksnmp>. Contact: </sparkbridge/contact>. Release downloads and checksums: [GitHub releases](#).